Analysis of Image Forgery Detection Using Canny Edge Detector

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***Abstract*— As now-a-days many image processing software and editing tools are available using which image can be easily faked. In various fields digital images are used as legal evidence, for forensics investigations,, so there is need of making such system that can detect image forgery. The passive approach of image forgery detection provides image authencity without having any information about the digital image. Further in this paper we have discussed various forgery techniques which were used for creating forgeries in the digital picture. We proposed an optimized method that can detect the forgery in the image using k-means and feature matching algorithm which uses canny edge detector. It gives recall between 85 to 95% and precision 100%.We run this system over two datasets.**

Keywords*— *SIFT, SLIC, K-means classification, copy-move forgery, canny edge detector.

I. INTRODUCTION

Forgery term is not new to humans but in 90s 20s it has become a huge problem that need to be taken into consideration. First cases were detected of forgeries but it were related to signatures, watermark but as the usage of internet [6] increased the digital image related forgeries case are increased. Due to editing tools digital images can be easily damaged which human cannot detect only by seeing [6]. So, there is need to develop techniques to validate integrity and authencity of the images, as this images are considered as the evidence in various fields like in investigation or in medical field as medical record or as financial documents, etc.

The major goals of this paper is:-

To make a system that detect forgery images.

To review pixel based forgery techniques.

To do comparative study of this existing techniques.

There are mainly five types of image forgery detection techniques as shown in fig 1 which come under two categories one is active and second one is passive from which active is not used mostly in real life its very rare to detect this type of forgery whereas passive is used mostly to tamper an image in real life which is used often to create and detect forgeries in day to day life.

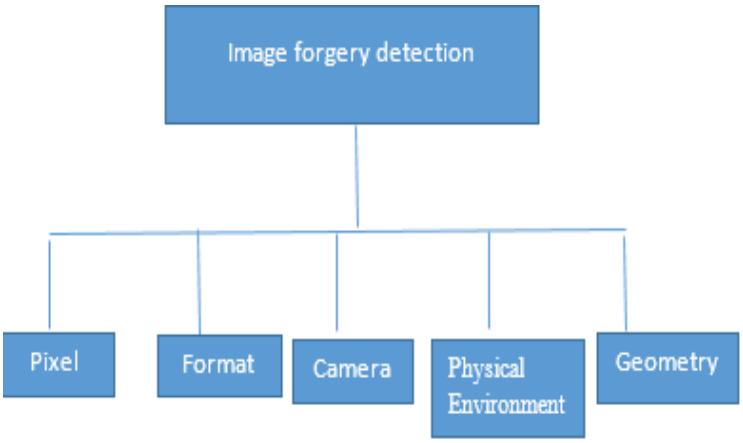

Figure 1 Image forgery Detection Techniques

1. Pixel Based Detection Technique:

Pixel based technique [6] mainly focuses on pixels of the images. It's mainly divided into 4 types as shown in fig 2.

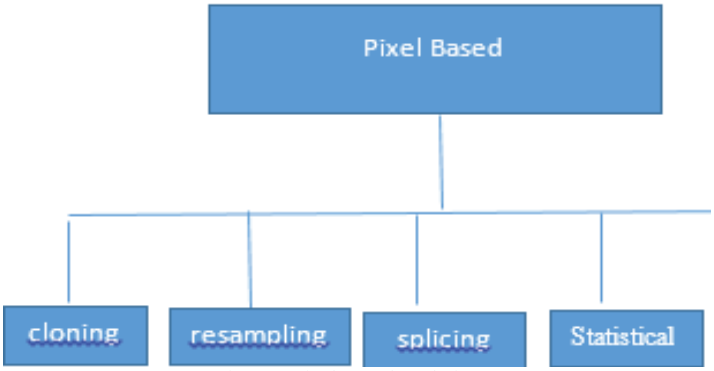

Figure 2 Pixel Based Technique

2. Format Based Detection Technique:

Format Based technique [6] works mainly on JPEG format. It detects forgery of a compressed image. It's divided into 3 types as shown in fig 3

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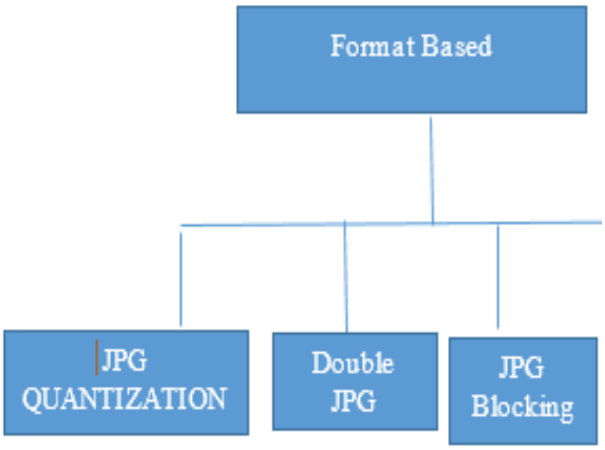

Figure 3 Format Based Technique

3. Camera Based Detection Technique:

While capturing the image it goes various steps from quantization to JPEG compression. This steps vary depending on the camera model. This camera based technique [6] works on this principle. It's mainly divided into 4 types as given in fig 4.

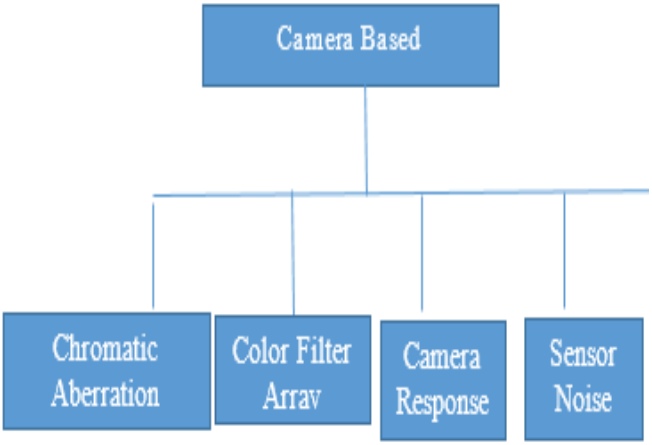

Figure 4 Camera Based Technique

4. Physical Environment Based Detection Technique:

While capturing an image lighting is most important. Physical environment based technique [6] works on the light based environment under this environment image is captured. This light can be considered to detect tampering is done in image or not. It is mainly divided into 3 types as shown in fig 5

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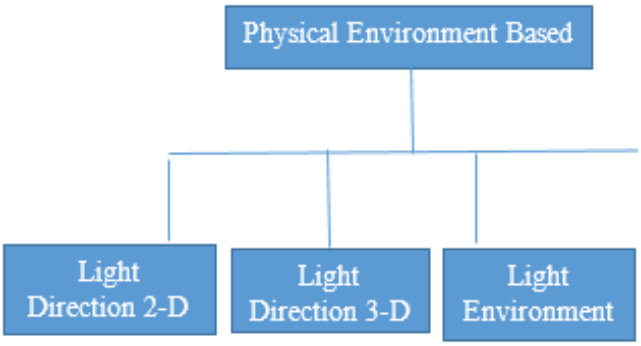

Figure 5 Physical Environment Based Technique

5. Geometry Based Detection Technique:

Geometry based technique [6] considers measurement of objects and position relative to camera. It's further divided into 2 types as shown in fig 6.

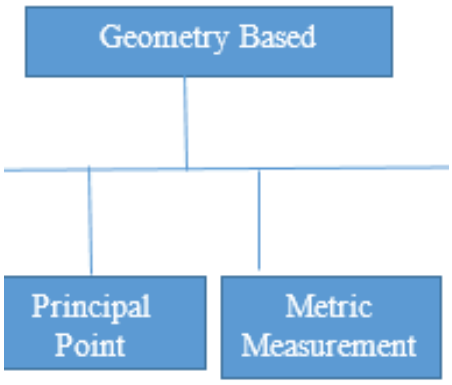

Figure 6 Geometry Based Technique

In remaining sections we are going to describe the literature review, implementation, result and discussion, conclusion, references.

II. Literature Review

Passive Based Detection technique as shown in the fig 2 is mostly use by people to tamper an image. Image tampering is nothing but do making changes, modifying image such that it looks like real image but in reality it's not the real image it's a fake image. The statistical [6] type is explained at the pixel level

A. Cloning Pixel Based Technique:

In cloning pixel based technique[6] a part of image is removed and pasted in some other part of the same picture.For example Fig 7a shows original image has six ballons but in another Fig 7b it shows nine ballons.

Figure 7a. Original Image

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Figure 7b. Copy move image

B. Resampling Pixel Based Technique:

In Resampling pixel based technique[7] original image is faked by changing its size,bu rotating and by scaling the image.For example a person can be duplicated by scaling and by increasing its weight and hieght as same as that original person.This process is known as resampling an image.

C. Splicing Pixel Based Technique:

In Splicing pixel based technique[8] two images are combined to form a single image.For example one image has beach and second image has sand castle so if we combine both the images into one image it will have both beach as well as sand castle in one image as shown in fig8a,8b,8c.Here 8c image is the splicied image.8a and 8b are two different images which combined together form 8c image.

Figure 8a. First Image

Figure 8b. Second Image

Figure 8c. Splicied Image

III. IMPLEMENTATION OF IMAGE FORGERY DETECTION USING CANNY EDGE SYSTEM

In this image forgery detection using canny edge system we use image for checking its authentication and whether it's tampered. In this we are implementing a system that detects more accurately the forged images.for web applications. We are applying various methods on the images through which we can decide whether it is tampered. First we need to input the image to check whether this image is manipulated .After browsing we apply segmentation on that image so that we can extract features from that image easily. After segmentation to extract features feature extraction algorithm is used. After feature extraction we have classified this features using unsupervised k-means algorithm. After classification we have found the edges using canny edge detector. After finding the edges we have matched that edges with the features we got. Through this we get the specific region that is tampered from the original image. Then we calculate the precision and recall of that image. The architecture diagram as shown in fig 9 of the image forgery detection using canny edge system is as follows:-

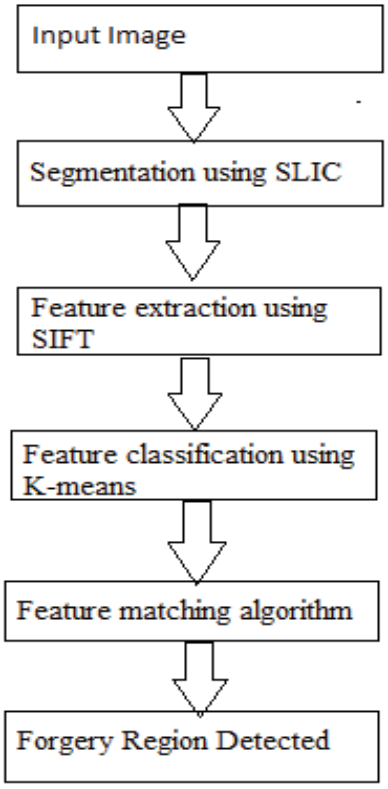

Figure 9. Architecture Diagram of image forgery detection using canny edge system

Components are explained as follows:-

1. **Segmentation:**-In this the original image is segmented using SLIC superpixel algorithm.

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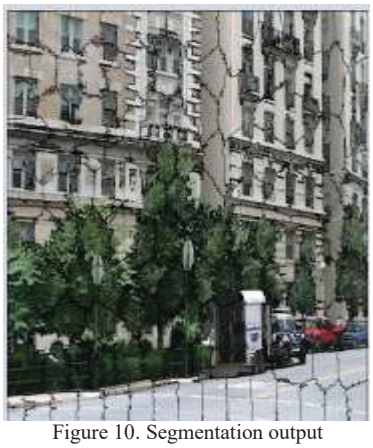

Figure 10. Segmentation output

2. **Feature Extraction**:-In this from segmented image the features are extracted using SIFT feature extraction algorithm.

Figure 11. Feature extraction output

3. **K-means**:-The feature extracted from image is classified using unsupervised algorithm.

K-means:-
Its unsupervised classification algorithm used to classify the features that are extracted from the features extracted through feature extraction method.

Algorithm is as follows:-

a. Randomly choose k objects from D as the initial cluster centroids.
b. For each of the objects in D do
c. Compute distance between the current objects and k cluster centroids
d. Assign the current object to that cluster to which it is closest.
e. Compute the "cluster centers" of each cluster. These become the new cluster centroids.
f. Repeat step until the convergence criterion is satisfied
g. Stop

4. **Canny detector**:-Through this edges are discovered of the image.

Canny Detector
The canny edge detector is used to identify the edges of the image.
Algorithm:

a. Apply a Gaussian blur on image
b. After that find edge gradient strength and direction
c. Do the discovering along the edges
d. Put out untold edges from the file and save that file in form of matrix object

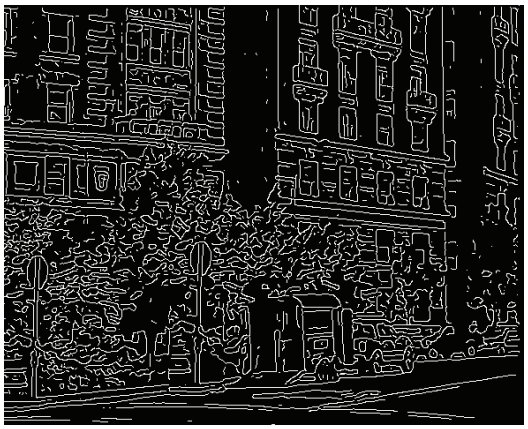

Figure 12. Canny output

5. **Matching Algorithm**:-Through matching algorithm we get the exact tampered area or region i.e. our main output of this project. In this for image forgery detection using canny edge system we have matched keypoint within the edges to get exact forged region whereas in SLIC system they have only matched the keypoints.

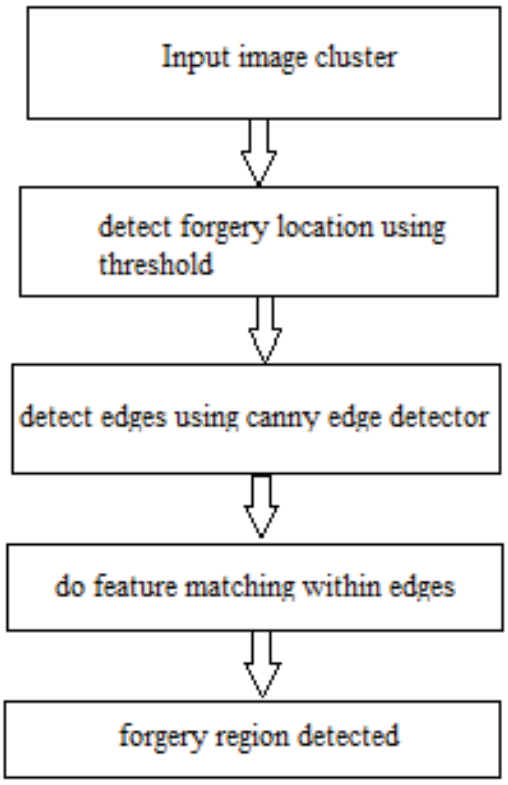

Figure 13. Flowchart of image forgery detection using canny edge system matching algorithm

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Matching Algorithm:
Matching algorithm used in our proposed system using canny detection is as follows and its flowchart is as shown in the fig 10.
Algorithm works as follows:

a. Detected the forgery locations using threshold values
b. Apply canny edge detector
c. Detect the edges of the given object
d. Then match the keypoints within the edges
e. After matching the keypoints within edges we get accurate forgery region which is the potential copy.
f. After that we have to match this forgery region with the ground truth image if both matches that means the image is tampered, calculate the score to see how accurately we have detected the forgery region.
g. If they both don't match then the given input image is original and no tampering is done on it.

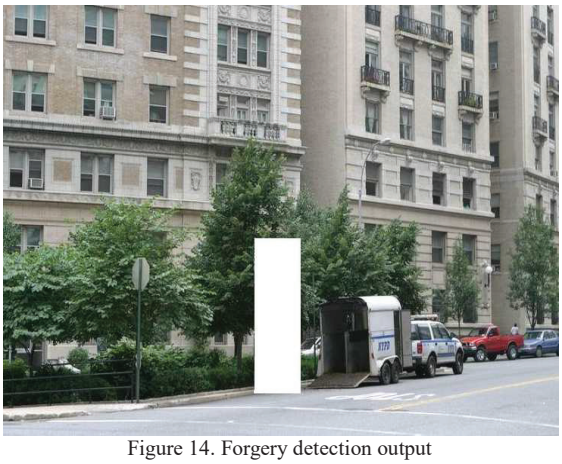


Figure 14. Forgery detection output

IV. Result and Discussion

Recall, Precision, F1, Accuracy average values are calculated for .image forgery detection using canny edge system on both the dataset.

A. Experimental Analysis

The SLIC and image forgery detection using canny edge system can detect the copy move forgery detection using superpixel SLIC algorithm for segmentation, feature extraction using SIFT. In image forgery detection using canny edge system feature matching is done within the edges, region extraction is also done. The image forgery detection using canny edge system working steps are as follows:

a. The system works on java platform, jdk 1, 8, Ide used is Netbeans. It requires 2.2GHz, 200GB hard disk, it requires 4GB RAM.
b. Data is extracted from MICC-F220 which has 220 images out of which 110 are modified and 110 are real images We got this dataset through http://lci.micc.unifi.it/labd/2015/01/copy-move-forgery-detection-and-localization/ this link and Mendeley dataset which has 100 original and 1050 tampered images and 100 mask images which we got through the below link. https://data.mendeley.com/datasets/dk84bmnyw9/2
c. Images are segmented using super-pixel SLIC algorithm. Features are extracted using SIFT from block-based images and feature points are matched within the edges using feature matching algorithm. The forged region is detected and extracted and later the scores are calculated.

B. Comparison Result

Recall, Precision, F1 score:-After finding the tampered region we find the recall, precision and F1 measure through k-means and for accuracy we need the ground truth image.

a. **Precision**:-Precision can be defined as how accurate is your model out of those predicted positive and how many of them are actual positive. It can be written as:

Precision=True Positive/Total predicted Positive.

b. **Recall**:-Recall calculates how many actual positives our model captured through labelling it as positive. It can be written as:

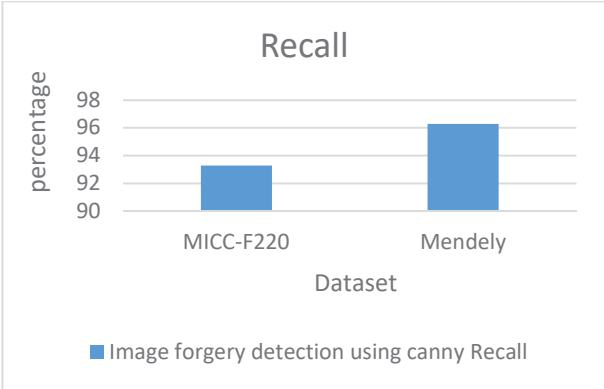
Recall=True Positive/Total actual Positive

c. **F1 score**:-It is a model metric which used to seek balance between recall and precision and when there is uneven class distribution.

F1=2*((Precision*Recall) / (Precision + Recall))

TABLE 1 Comparison of RECALL between image forgery detection system and without canny system (existing system)

Column1	MICC-F220	Mendely
Image forgery detection using canny Recall	93.29	96.278



Bar Chart 1. Shows comparison of average recall values when image forgery detection using canny edge system and without canny system was applied on two datasets

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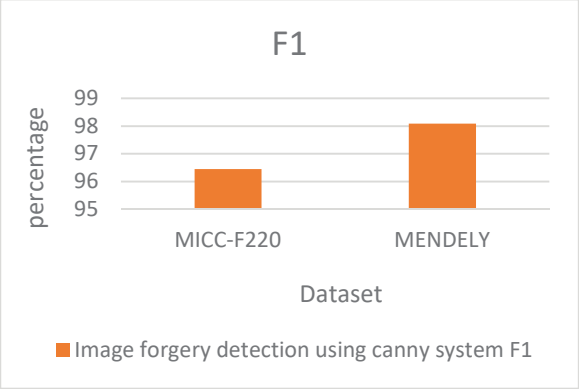
TABLE 2. Comparison of F1 between image forgery detection system and without canny system (existing system)

Column1	MICC-F220	MENDELY
Image forgery detection using canny system F1	96.45	98.09



Bar chart 2 Shows comparison of average F1 values when image forgery detection using canny edge system and without canny system was applied on two datasets

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C. *Confusion Matrix*

Confusion matrix is performance to make decision that can be made by classifier.

Table 3. Confusion Matrix of image forgery using canny edge system

Canny System	Predicted		
		Background	Forged
True	Background	0	10675
	Forged	0	237

D. *Comparative Analysis*

Table 4 Comparative Analysis with Existing System

Sr.No	System	Accuracy
1	SWT[1]	93%
2	LSH using k-means[2]	72.74
3	LSH[3]	67.38
4	FCM[4]	71
5	Image forgery detection using canny edge system	96.27

V. CONCLUSION

We conclude that our image forgery detection model based on new matching algorithm that can detect the forgery regions if any in the given image. It helps to check the authencity of the image. It can be used in various fields to check the image which are presented as a proof. It gives recall between 80 to 95% and precision up to 100% and the F1 measure between 85 to 97%.It's very efficient, low cost and easy to use and detects forgery area more accurate as compared to other existing systems.

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